ZCAS University

CIT4361 DIGITAL FORENSICS

FINAL EXAMINATION

TUESDAY 2nd JUNE 2026

09.00 – 12.00 HOURS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: Question One in Section A is **compulsory**.
2. Sections B: Answer Three (3) questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

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SECTION A: QUESTION 1 IS COMPULSORY AND MUST BE ATTEMPTED

Question 1:

Scenario: Read the scenario in Appendix A (also sent to you earlier), and then answer the following questions:

a) The NIST guidelines list six types of mobile forensics methods that could have been used to extract mobile device evidential data by investigators at ZamNet Financial Services Ltd. Explain the said mobile device forensic acquisition methods. (10 Marks)

b) The evolution in computing from the Internet of Thing (IoT) to Internet of Everything (IoE) to Internet of Anything (IoA) has given rise to forensics challenges in mobile device investigations. Discuss IoT forensic challenges in mobile device investigations. (10 Marks)

c) Mobile devices are, unlike before now capable of using cloud computing facilities. Explain cloud synchronization artifacts in digital forensics. (10 Marks)

d) Messages might still be received on a mobile device after it has been seized for investigations. Discuss the actions you should undertake to stop messages being received and outline the recommendations given by SANS DFIR when faced with such a situation. (10 Marks)

[Total: 40 Marks]

SECTION B: ATTEMPT ONLY THREE QUESTIONS FROM THIS SECTION

Question 2:

a) As a Forensic Investigator for ZamNet Financial Services Ltd. Discuss the "Rules of Evidence" that you would be expected to follow during your forensic investigations. (10 Marks)

b) Based on the evidence adduced from your investigations, decide whether this should be treated as a criminal or civil case. Give reasons. (6 Marks)

c) Explain the importance of securing a digital crime scene during an investigation. (4 Marks)

[Total: 20 Marks]

Question 3:

a) Explain the role of the Windows Registry in forensic investigations. (10 Marks)

b) Explain how deleted NTFS files can be recovered during a forensic investigation. (5 Marks)

c) Discuss APFS forensic artifacts on Macintosh systems. (5 Marks)

[Total: 20 Marks]

Question 4:

a) Hashing is used in digital forensics to ensure the integrity of collected evidential data. Distinguish Cyclic Redundancy Check (CRC), Message Digest 5 (MD5) and Secure Hash Algorithm version 1 (SHA-1) hash techniques and outline the three rules of forensic hashes. (10 Marks)

b) Data hiding was used in the ZamNet Financial Services Ltd. Case. Discuss five data hiding techniques that can be used to hide evidence. (5 Marks)

c) Distinguish Data Hiding and Steganography. Discuss the steganalysis attack methods that can be used in steganalysis. (5 Marks)

[Total: 20 Marks]

Question 5:

a) Explain the importance of e-mail headers in forensic investigations. (10 Marks

b) Explain legal considerations in e-mail investigations. (5 Marks)

c) Discuss the various elements involved in social media forensic analysis. (5 Marks)

[Total: 20 Marks]

APPENDIX A

Case Study: Cyberattack Investigation at ZamNet Financial Services Ltd.

Background

ZamNet Financial Services Ltd. is a medium-sized financial technology company based in Lusaka that provides mobile banking, online payment processing, and cloud-based financial services to customers across Southern Africa. The company maintains Windows-based workstations for employees, Linux-based web servers, and a Bring Your Own Device (BYOD) policy for executives and field staff.

In January 2026, the organization experienced a major cybersecurity incident involving unauthorized access to customer financial records, suspicious e-mail communications, and data leakage through social media platforms. The incident began when employees received phishing e-mails pretending to originate from the company's IT support department. Several users clicked malicious links and unknowingly installed malware onto their Windows systems.

A week later, the company's Security Operations Center (SOC) detected unusual outbound network traffic from one of the Linux web servers hosting customer databases. Investigators also discovered that sensitive customer information had been posted anonymously on a social media platform. Simultaneously, one executive reported that his smartphone had been stolen during an international business trip, raising concerns about unauthorized access to company e-mails and confidential files stored on the device.

The company immediately initiated a digital forensic investigation team consisting of forensic analysts, incident responders, legal officers, and law enforcement representatives.

Processing Crime and Incident Scenes

Incident Scene Processing

The forensic team arrived at the organization's headquarters and secured the digital crime scene. Investigators documented the physical environment using photographs, sketches, and evidence logs. All potentially affected devices were identified, including desktop computers, laptops, routers, servers, external hard drives, and mobile devices.

Investigators followed the chain of custody procedures to ensure all evidence remained admissible in court. Devices were labelled, packaged in anti-static bags, and transported to the forensic laboratory.

Volatile data such as RAM contents, active network sessions, and running processes were collected before shutting down compromised systems. Investigators also interviewed employees to determine who had access to the affected systems and when suspicious activities first occurred.

Working with Windows and CLI Systems

Windows and Command-Line Investigation

Forensic analysts examined compromised Windows workstations using forensic imaging tools. Bit-stream copies of hard drives were created to preserve original evidence integrity.

Using Windows Event Viewer and command-line tools such as:

- *ipconfig*
- *netstat*
- *tasklist*
- *systeminfo*
- *wmic*

investigators discovered evidence of unauthorized remote access sessions and malware persistence mechanisms.

The Windows Registry revealed suspicious startup entries that automatically launched malicious software during system boot. Deleted files were recovered from NTFS partitions, including spreadsheets containing customer financial records.

Investigators also analyzed PowerShell logs and command history files, which showed attackers using scripts to escalate privileges and exfiltrate data.

Linux and Macintosh File Systems

Linux Server Investigation

The compromised Linux web server used the ext4 file system. Investigators examined log files located in directories such as:

- */var/log/auth.log*
- */var/log/syslog*
- */var/log/apache2/access.log*

Analysis revealed repeated brute-force login attempts followed by successful SSH authentication from a foreign IP address.

File system metadata analysis identified unauthorized modifications to web application files. Investigators discovered hidden directories containing compressed archives of stolen customer records.

A Macintosh laptop belonging to the marketing manager was also analyzed. The APFS file system contained browser artifacts, cached credentials, and social media login tokens that attackers used to publish leaked information online.

Digital Forensics Analysis and Validation

Evidence Analysis and Validation

The forensic laboratory validated all forensic images using MD5 and SHA-256 hash values to confirm evidence integrity. Investigators correlated timestamps from Windows systems, Linux servers, mobile devices, and network logs to reconstruct the attack timeline.

Specialized forensic tools were used to analyze malware behavior, browser history, deleted files, and network captures. Analysts verified findings through repeatable testing procedures and peer review to reduce errors and ensure reliability.

The investigation concluded that attackers initially gained access through phishing e-mails, then escalated privileges, accessed customer databases, and leaked information through social media channels.

E-mail and Social Media Investigations

E-mail and Social Media Evidence

Investigators analyzed suspicious e-mails by examining:

- E-mail headers
- SMTP routing information
- IP addresses
- Attachments
- Embedded URLs

The phishing e-mails originated from spoofed domains resembling the company's legitimate domain name.

Social media forensic analysis revealed that attackers used compromised employee credentials to post confidential customer information. Metadata from uploaded files and timestamps linked the activity to the compromised Macintosh laptop.

Chat logs and direct messages collected under legal authorization further connected suspects to the attack.

Mobile Device Forensics and the Internet of Anything

Mobile Device and IoT Investigation

The stolen executive smartphone contained company e-mails, VPN credentials, and authentication tokens. Investigators used mobile forensic tools to extract:

- SMS messages
- Call logs
- GPS location data
- Application data
- Cloud synchronization records

Analysis showed that the device had connected to public Wi-Fi networks shortly before suspicious logins occurred.

The company also used IoT-enabled smart security cameras and biometric access systems. Investigators examined IoT logs and discovered that attackers had attempted to disable certain security devices during the intrusion.

The investigation ultimately resulted in the identification of insider collaboration with external cybercriminals.